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The Internet of toys

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ABSTRACT

The Internet of Toys refers to a future where toys not only relate one-on-one to children but are wirelessly connected to other toys and/or database data. While existing toy companies and start-ups are eagerly innovating in this area, problems involving data hacking and other privacy issues have already occurred. The Hello Barbie and VTech hacks in late 2015 are recent examples. This article reviews, outlines, and analyses these recent advances in children's engagement with the Internet. It shows how Internet-connected toys, among other data-inducing practices (such as baby wearables and school analytics), are implicated in big data processes that are datafying a generation of youngsters. Significant issues exist around the data security and safety of The Internet of Toys for child consumers who are usually too young to fully understand and consent to data collection or to understand other security issues.

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Introduction

The Internet of Toys is derived from the Internet of Things (IoT) and this refers to 'a future in which digital and physical entities can be linked, by means of appropriate information and communication technologies, to enable a whole new class of applications and services' (Miorandi, Sicari, De Pellegrini, & Chlamtac, 2012, p. 1497). Although visionary when the term was first coined in 1999, the IoT is already operational with smartphone apps that remotely control home-based objects from distant locations, as well as wearable technology measuring owners' sleep patterns, heart rate, and exercise regimes (Sung, 2015).

Internet-connected toys build on these technological successes. Such toys include: Hello Barbie and Smart Toy Bear, which use voice and/or image recognition, connecting to the cloud to analyse, process, and respond to children's conversations and images. Other examples are: app-enabled toys such as drones, toy cars and robots (Star Wars, BB-8 Droid); toys-to-life, which connect action figures to video games (Skylanders, Amiibo); puzzle and building games (Osmo, Lego Fusion); and children's tech wearables such as smart watches and fitness trackers.

The emerging risks of Internet-connected toys to children and families include corporate and government surveillance of children's activities and encroachments

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upon their data privacy and security. The use of IT technologies to monitor individual children's online activities already occurs at a corporate level and will expand further with Internet-connected toys. Other risks include hacked surveillance of Internet-connected toys, geo-locational tracking of children and remote control of toys' various recording and 'speaking' technologies by others. Calamitous security failures already include the hacking of Mattel's Wi-Fi Hello Barbie (Nov, 2015) and VTech's Learning Lodge (Nov, 2015).

Toy companies are not taking responsibility for these privacy and security failures. Instead they're using complicated terms and conditions to shift responsibility to consumers – and in the case of children to their parents. For example, VTech recently changed its terms and conditions to give notice that it is not responsible for any breach of data (Bogart, 2016). The safety and security of children's data requires companies to go beyond the 'opt-in-opt-out' paradigm of data privacy. Policy and practice should interrogate alternative means of addressing the privacy of children and ethical issues raised by their innocent participation in the Internet of Toys (IoToys).

This article offers a critical analysis of children's technology and media use. The term 'critical' identifies a specific focus on explicit and implicit power relations and acknowledges the role of the global communications industry 'dominated by commercial interests' (Mizen, Hutchby, Pole, Moran-Ellis, & Bolton, 2001, p. x). The paper adopts a children's rights framework, based on post-digital conceptions of the 1989 United Nations' *Convention on the Rights of the Child* (CRC) (Livingstone & Bulger, 2014). The critical focus is upon commercial interests and the political economy of Internet-connected toys, emphasising children's rights to participation, and their rights to protection, elements of Livingstone and O'Neill's 'protection, provision, and participation' approach (2014). As Hamelink and Hoffman argue (2008, p. 14), 'the protection of private life [and children's rights, we suggest] comes with positive obligations that go beyond non-interference. International courts have stressed that the advance in technology and therewith the potential for surveillance of citizens must be equalled with rules to prevent an erosion of rights'.

The paper adopts a case study approach (Yin, 2014) to analyse and identify consumer concerns in relation to the Internet of Toys, as shaped by news and other public discourses that describe and evaluate the risks and benefits to children and family life of these products. Notwithstanding the possibility of the social amplification of risk via news media and specialist commentary (Thiesse, 2007), the media draws consumer risks and concerns to public attention, playing a 'significant role in forming and shaping risk scenarios' (Slettemeås, 2009, p. 230–1). The security failures and consumer anxieties described in the following case studies deliver critical counter narratives to the utopian discourses promoted by the commercial interests promoting Internet-connected toys, raising crucial questions about children's rights.

Current literature regarding children's screen-based Internet use indicates ongoing privacy and security concerns regarding the collection of children's personal information and opinions. However, little research has been published in the area of Internet-connected toys, apart from technical articles outlining software and hardware developments (Corbellini, Aksit, Schmid, Mangold, & Gross, 2014; Fan, Shin, & Choudhury, 2014). There has, however, been an increasing volume of social commentary and news

items about these toys, as well as technical commentary about security flaws as these emerge (Adhikari, 2015a; Emm, 2015; Kube, 2015), discussed below.

Concerns raised include the lack of regulatory direction relating to children's data privacy and security within this market niche. The complex networked nature of IoT technologies will most likely involve currently unforeseen data security and privacy risks emerging over time, since it is not currently possible to define all the possible concerns that may emerge. The authors argue that, in light of spectacular security failures that have already occurred, policy statements and initiatives that combine effective direction to industry with appropriate sanctions should be researched and instigated.

Relevant literature

Children, media, and technology

The relationship between children and media is traditionally conceptualised as one of consuming media content, but contemporary communication technologies afford a more dynamic and individualised media experience for children. A rights based approach argues for children's legitimate participation in the digital public sphere, acknowledging that children are 'seeking content on their own, posting and sharing it themselves, and interacting with mobile apps and other digital platforms' (Montgomery, 2015, p. 270). This introduces a complexity and fluidity to children's relationships with media opening them up to sophisticated data collection practices at an early age.

The Internet and digital technologies make conventional boundaries between the private and public sphere less distinct. Thus, children's presence in the digital realm challenges the notion of childhood as being separate and protected from the wider world. As Grimes puts it: 'Whereas children have historically been excluded from directly participating in most aspects of the public sphere – from voting to mass media production – the Internet now offers users of all ages and ability a variety of opportunities for participation' (2008, pp. 66–7). Teens and children, in particular, have taken the opportunity to engage in the digital public sphere with great enthusiasm. At the same time, this enthusiasm brings with it a number of challenges. Grimes suggests that:

> Every day, children enter into complex relationships with the adults who create and manage profit-driven online spaces directed at kids. These include business relationships, consumer relationships, and legal relationships, most of which occur 'behind-the-screen' of seemingly entertainment oriented spaces (2008, p. 67).

Hence, while children enjoy hours of online entertainment and social interaction within communities of interest, exercising their CRC 'participation' rights of freedom of expression (Art. 13), freedom of association (Art. 15), and engagement in cultural life (Art. 31); a large function of these sites is to provide opportunities for branding and market research, processing and packaging children's data. These are invariably sites where children's personal information and opinions are gathered (Kapur, 1999; Montgomery, 2015; Steeves, 2006), potentially infringing CRC Art. 16 which protects

children from 'arbitrary or unlawful interference with his or her privacy, family, or correspondence' (Livingstone, 2015, p. 5).

Children's privacy protection to date

In 2000, the US Federal Trade Commission inaugurated the Children's Online Privacy Protection Act (COPPA) to deal with the specific issue of collecting personal information relating to children under 13. The COPPA Act obliges websites or online services to, among other things, 'provide notice that they are collecting children's personal information, and collect verifiable parental consent' (COPPA 2000; cited in Holloway, 2014, p. 7). This is one reason why many online services (for example Facebook) have 13 as the minimum age for participation.

Australian regulation does not specifically address children's information privacy and security, even though new Australian Privacy Principles (APP) came into force in 2014 (Office of the Australian Information Commissioner, 2014). The privacy principles recommend:

> 1.3 At or before the time (or, if that is not practicable, as soon as practicable after) an organisation collects personal information about an individual from the individual, the organisation must take reasonable steps to ensure that the individual is aware of:
>
> (a) the identity of the organisation and how to contact it; and
> (b) the fact that he or she is able to gain access to the information; and
> (c) the purposes for which the information is collected; and
> (d) the organisations (or the types of organisations) to which the organisation usually discloses information of that kind; and
> (e) any law that requires the particular information to be collected; and
> (f) the main consequences (if any) for the individual if all or part of the information is not provided. (Office of the Australian Information Commissioner, 2014)

Compliance with the APP and COPPA requirements is typically covered through users' consent to privacy policies and permissions within children's apps and websites. However, the capacity of children (and their parents) to evaluate and consent to the legalise packaging of privacy policies and permissions is questionable. The use of multiple devices with different operating systems complicates the security and safety of children's Internet use (Holloway, 2013, p. 19) while Terms of Service contracts are often hard to locate and consist 'of lengthy, indecipherable texts (ranging between 3 and 12 single-spaced printed pages)' (Grimes, 2008, p. 75). Most parents and children scroll through these consent processes, which challenge the notion of fair practice in terms of market research ethics and user rights. Further, a 2012 US Federal Trade Commission report found that privacy disclosure and information collection and sharing practices tended to be compromised within children's apps. Their research indicated that: 'nearly 60% (235) of the apps reviewed transmitted device ID to the developer or, more commonly, an advertising network, analytics company, or other third party [… while] only 20% (81) of the apps reviewed disclosed any information about the app's privacy practices' (Mohapatra & Hasty, 2012, p. 6).

Moreover, those companies that do disclose their data collecting practices warn users that their 'user experience' will be restricted, or access banned altogether, if consumers

opt out of data collecting processes. For instance Club Penguin, a well love children's virtual world, has a comprehensive privacy policy with full disclosure of the wide-ranging information that Disney collects. This includes users' location, all text content in children's messages and posts to public forums, all transaction and registration information, and any information 'you provide to us when you use our sites and applications on third party sites or platforms' (The Walt Disney Company, 2013). This data can be shared within the 'Walt Disney Family of Companies' and with other third parties (The Walt Disney Company, 2013). Children or their parents are given a choice to opt out of these information collection and sharing practices. If they do opt out, however, they have a sub-optimal playing experience: 'Please be aware that, if you do not allow us to collect personal information from you, we may not be able to deliver certain products and services to you, and some of our services may not be able to take account of your interests and preferences' (The Walt Disney Company, 2013).

In another example, the terms and conditions of Hello Barbie's companion app essentially blocks users from accessing the service if they do not agree to all terms and conditions. These service conditions involve extensive data collection including speech data capture, storage, analysis, and the sharing of all the child's digital interactions with their Hello Barbie (Toytalk, 2016). Toytalk, the company with which Mattel partnered for the design and service of Hello Barbie's companion app, states that 'If you do not agree to all of the terms and conditions of the terms, Toytalk is unwilling to grant you (or your children) the right to use the companion apps and the services' (Hello Barbie™/Barbie Hello Dreamhouse™ Companion App Terms of Use, 2016).

The 'opt-in-opt-out' choices presented to parents entice them to agree to all the terms and conditions in order to fully realise the affordances of their children's digital games and toys. This positions parents and caregivers as having sole responsibility for their children's data privacy and security by transferring legal responsibility for the collection, storage, analysis, and sharing of children's data to parents. These strategies give commercial entities a relatively free hand to continue, and perhaps expand upon, their data collecting practices, even though the children are under 13. One impact is the potential infringement of CRC Art. 12 which notes the right of children 'to be consulted in all matters affecting them' (Livingstone, 2015, p. 5).

Potential benefits of Internet-connected toys

Internet-connected toys and games offer a range of potential benefits to children, apart from the obvious enthusiasm and enjoyment youngsters experience, and despite the worrying issues relating to children's data security and privacy. The personalisation of interactive media, which analyses and responds to children's individual input, offers significant educational benefits. Individualisation is at the core of dramatic changes to learning technologies which use software that adapts to an individual child's needs and progress, 'giving students choice in the pace, place, and mode of their learning' (Gordon, 2014, p. 3). These refinements create a flexible learning 'environment for children, where the lines between formal and informal learning are being blurred' (Montgomery, 2015, p. 268). Internet-connected toys can also make programming and 3D design more accessible for children. For instance, the app-enabled robotic ball that forms the basis of Star Wars' BB-8 Droid (called Sphero) houses the SPRK

program which allows users to 'invent their own uses for the app-controlled sphere' (Irwin, 2016). Mattel's ThinkMaker, an app-connected 3D printer with which children can create their own toys and jewellery, also makes the learning of 3D design and manufacture available to children (Barrett, 2016).

With toy industry leaders such as Lego, Mattel, and Vivid investing heavily in research and development (The LEGO Foundation; Mattel Research and Development; Vivid Imaginations), sales of Internet-connected toys are expected to grow rapidly. These developments promise each child a customised, distinctive, and dynamic interaction with their toys using the Internet, in line with their CRC rights to participate in cultural life (above). Questions about the integration of these new play-things into children's culture are yet to be answered. Such questions include: whether or not these toys facilitate existing play patterns, or change the ways in which children play with toys; whether these toys allow for open-ended creative play; and, whether these toys foster particular dependencies or vulnerabilities which impact upon a child's worldview and mode of life. At the same time, the personalisation that characterises Internet-connected toys aligns with children's CRC 'provision' rights for 'an education to support the development of their full potential (Art. 28)', 'to recreation and leisure appropriate to their age (Art. 31)' and, potentially, 'to diverse material of social and cultural benefit to the child (including minorities) to promote children's well-being (Art. 17)' (Livingstone, 2015, p. 5).

Case study: hacking the Internet of toys

Research into children's relationship with media sometimes overlooks the role of commercial forces and how business priorities promote the generation and intensification of user interaction – and the production of big data. Internet-connected toys are the newest data-driven media platform to collect information about children's activities, interactions, preferences, and behaviours. The case studies (Yin, 2014) outlined below describe data security failures of two early entrants into the field of Internet-connected toys.

Hello Barbie is the most recent generation of a Mattel toy that has been in the market since 1959. According to Ruth Handler, Barbie's creator, 'through the doll, the little girl could be anything she wanted to be. Barbie always represented the fact that a woman has choices' (Mattel, n.d.a). Generally regarded as a role model for a Kardashian-like *fashion-ista* rather than Rosie the Riveter, Barbie has prompted extensive commentary and concern regarding her supposed reinforcement of gendered, racial and sexual(ised) stereotypes (Rush & La Nauze, 2006; Schoppmeier, 2015; Sherman & Zurbriggen, 2014), and her promotion of an unreal body image (Dittmar, Halliwell, & Ive, 2006). Even so, the doll remains a global market leader with Mattel claiming that 'since 2011, the brand has consistently produced worldwide annual gross sales in excess of $1 billion' (Mattel, n.d.b).

Barbie's growth is not a given, however, and concerned about a recent downturn in sales in some mature markets, Mattel has recently embraced a vision of a more diverse Barbie, announcing that 'curvy, petite, and tall' versions of Barbie will soon become available (Ziobro, 2016). These innovations follow on from a 2015 announcement that Mattel was partnering with cloud-computing specialist ToyTalk to provide an interactive version of Barbie 'that converses with kids, remembers things they say, and

recalls details later' (Newman, 2015). Mattel's subsequent launch of Wi-Fi-enabled Hello Barbie in the US was against the backdrop of fighting to regain market share in mature consumer markets.

Security specialist Matthew Jakubowski has hacked Hello Barbie to reveal her technological vulnerabilities and demonstrated that not everything is rosy in the techno-toy cupboard. According to Adhikari (2015b), Jakubowski's hack of Hello Barbie's operating system collected data including the names of Wi-Fi networks, and identified details of accounts and the sound files used for Barbie's default pre-recorded responses (for use when the toy is not connected to the cloud). When fully operational, and with her belt buckle pressed, Hello Barbie connects to ToyTalk's cloud servers and operates rather like Apple's Siri, using voice recognition and artificial intelligence to craft personalised responses. Newsweek's Lauren Walker (2015) describes how 'Barbie builds a customised cloud-based database of her owner's likes and dislikes, which she's then able to incorporate into conversation'. Angela Campbell, of Georgetown University's Center on Privacy and Technology, notes that 'If I had a young child, I would be very concerned that my child's intimate conversations with her doll were being recorded and analysed' (Newman, 2015), while Adhikari (2015b) cites Jakubowski as saying that the information 'could be used to access the home Wi-Fi network of the doll's owner and everything Hello Barbie records'.

Stilgherrin, an Australian security expert, perceives an issue in that toy companies are the driving force behind the Internet of Toys. 'We're going to see more and more of this kind of security vulnerability crop up', he said, 'simply because we're getting companies, in this case it's a toy manufacturer, [and] none of these organisations have a tradition or have the internal company infrastructure to constantly test and retest their devices and issue security updates' (Timms, 2015). ToyTalk's Martin Reddy speaks on behalf of the technological element of the Mattel–ToyTalk partnership and argues that his company has 'added client certificate authentication, above and beyond what most Internet-connected devices do, as a way to deter a casual attacker'. Jakubowski has also admitted that Hello Barbie features 'good levels of security that you don't typically see in many IoT devices. ToyTalk has certainly taken many of the concerns and has addressed them as best as they could' (Bogart, 2015). ToyTalk's view is that the risk is only at the point where the doll's user connects the toy to their Wi-Fi network. In Reddy's opinion, even if an attacker could circumvent the improved security they gain 'no access to Wi-Fi passwords, no access to child audio data, and cannot change what the doll says' (Reddy, cited in Adhikari, 2015b).

David Emm, principal security researcher with Kaspersky Lab, believes that 'Concerns about the doll centre mainly around privacy – the fact that secrets entrusted to the doll by a child are shared with Mattel and its partners. There's also the potential risk that such data might fall into the hands of hackers, if the security of Mattel or its partners are [*sic*] breached' (Emm, 2015). ToyTalk admits that this may be an issue and have addressed the concern in a company statement: 'We cannot prevent children from providing personal information when they talk with Hello Barbie, and such information may be captured in the recordings. However, it is our policy to delete such personal information where we become aware of it' (Bogart, 2015). This security level depends partly upon parental choice. Walker (2015) notes that 'If a parent chooses, the company will create and store audio files of the child–doll conversations on its website'. Although

parents would have to log into the site to access their child's audio files, such practices introduce a further risk, according to Adhikari (2015b). 'If hackers were to figure out [parents'] passwords, they could access the files as well'.

Stilgherrin explains the challenge by drawing an analogy with an existing privacy challenge which is ignored by many everyday technology users: 'If we're having trouble convincing people they need to install the security updates on their phone, it's going to be a lot harder to convince them to update the software on their doll' (Timms, 2015). In the face of industry and consumer concerns, however, Mattel's spokeswoman Marissa Beck argues that 'in all claims we know about, no children's audio files were accessed; no passwords were compromised; no personal information was disclosed; and no dolls were made to say anything unintended' (Adhikari, 2015b). Further, apparently impressed by Jakubowski's work, ToyTalk has announced a '"bug bounty" program that will reward security researchers for responsibly disclosing security vulnerabilities' (Bogart, 2015).

The confidence around the 'known hacks' of Hello Barbie would seem better founded if there was not a good reason to believe that Internet-connected technologies are risky. Unfortunately, this is far from the case. As Newman (2015) notes: 'In November 2014, British toymaker Vivid Toys debuted an Internet-connected doll, My Friend Cayla, that used speech recognition and artificial intelligence techniques to have a conversation with kids. By February, researchers had hacked the doll to spew curse words'. (Also see Walker, 2015.) Worse was to follow – particularly in terms of the November 2015 'data breach' (Newman, 2015) of Hong Kong company VTech's Internet-enabled Learning Lodge. Helpfully, VTech has itself commented at length on the issues it faced and is facing, while the company continues to update the Frequently Asked Questions relating to the hack (VTech, 2016).

According to the company, between 12 and 29 November 2015 'an unauthorised party accessed VTech customer data on our Learning Lodge app store customer database, the PlanetVTech and V.Smile Link websites and Kid Connect servers'. The services compromised include systems that allow 'children and parents to exchange voice and text messages, photos, drawings, and fun stickers' (VTech, 2016, Q9). On 29 November, 6 days after a journalist told them about the hack (on 23 November), VTech suspended Learning Lodge, the Kid Connect service and 13 other VTech websites while undertaking a security assessment.

As their investigation progressed, VTech discovered that the data breach affected 4,863,209 parents and 6,368,509 children on Learning Lodge, and approximately 1.2 million of these children had also enabled the Kid Connect service. Children's profiles on Kid Connect include 'name, gender, and birthdate' and, in some cases where parents have used a street address as their mailing address, these can be linked with children's photos and residential location, as well as to any voice and text messages between parents and kids. This is the case for all the Learning Lodge clients that had enabled Kid Connect, even though VTech emphasises that on Learning Lodge itself, 'kids' profiles only include name, gender, and birthdate' (VTech, 2016, Q13a, original emphasis). In addition to the Learning Lodge and Kid Connect hacks, a further 235,708 parent accounts, and 227,705 children's profiles were impacted on PlanetVTech and V. Smile Link (VTech, 2016, Q13b).

The locations of children affected by the VTech breach include US, Canada, France, Germany, Netherlands, Spain, UK, Australia, New Zealand, Hong Kong, 'and other

countries and territories' (VTech, 2016, Q22). Overall, VTech estimates that 18,171 Australian parents and 23,096 Australian children were impacted (VTech, 2016, Q14). In addition to details about the children involved, the hacked Learning Lodge account information included names, email addresses, and passwords for the parents' accounts, including secret questions and answers for password retrieval, IP address, mailing address, and download history (from the Learning Lodge store) (VTech, 2016, Q17). VTech imply that they take some comfort from the fact that VTech's databases do not contain credit or debit card information since 'our customers are directed to a secure third party gateway' (VTech, 2016, Q18). While Kid Connect has been relaunched in some countries, and there are plans to fully restore the service, PlanetVTech and the 13 suspended websites remain closed and VTech does not plan to reopen these services (VTech, 2016, Qs 7 & 8).

According to Josh Golin, executive director of the Campaign for a Commercial-Free Childhood, 'Trusting the companies to protect kids will not work […]. We absolutely need policy solutions to ensure these devices are secure' (Adhikari, 2015b). Australia's Office of the Children's eSafety Commissioner has recommended to parents that they should 'explain the situation to your children and advise them to be extra careful when going online. Make sure they know to come to you for advice if a website or application is asking for personal information' (Office of Children's eSafety Commissioner, 2015a). This is good advice, but will not make much difference when openly collected information, provided by parents, has been illegally hacked and 'hundreds of thousands of children and millions of parents [are] at risk of identity theft, fraud, or worse' (Kymberlee Price, senior director of research operations for Bugworld, cited in Adhikari, 2015b). Newman (2015) notes that 'Adults have to make their own choices about whether to trust tech companies with their data, but kids trust adults implicitly to make good cybersecurity decisions for them'.

Regulatory environment for Internet-connected toys

Toy safety standards in Australia are quite exacting. Both federal and state governments oversee the general safety of children's toys and the Australian Toy Association, which represents over 90% of manufacturers and retailers of toys within Australia, has developed and implemented the Australian Toy Standard (AS/NZ 8124). The Australian Toy Association works closely with the various state and federal regulators to ensure the safety and quick recall of problematic products (Australian Toy Association, 2015). To date, this year has seen hover boards, archery sets, floaties, drum sets, toy garages, and nightlights recalled in Australia due to a variety of physical safety problems ranging from fire hazards (hover boards) to swallowing hazards (drum set) (Australian Competition and Consumer Protection, 2016). The rigorous toys standards that Australia and many other countries have implemented, have been informed by the International Organisation for Standardisation which released its standards document International Toy Standard ISO 8124–1 in 2000, in a coordinated effort to standardise toy safety standards across countries (ATA, 2015).

Despite, or possibly because of, this fast moving market, it seems that toy safety bodies have paid little or no attention to the issue of Internet security for children's

toys. At the same time, communication regulators have not made the grade in terms of policy attention or development in this area. Positive exceptions are the recent Office of the Children's eSafety Commissioner, and Australia's and New Zealand's Office of Privacy Commission statements about security breaches associated with Internet-connected toys (Henwood, 2015; Office of Children's eSafety Commissioner, 2015b). Nonetheless, the question as to whether data safety and security concerns should come under regulatory and policy attention from toy standards bodies and/or communications authorities has not been publicly aired or resolved. This is a concern given that the Internet of Toys impacts specifically upon children who are too young to provide informed consent.

Notwithstanding the silence around the Internet of Toys, there are policy discussion papers and reports that focus on the IoT (Australian Communications and Media Authority, 2015). These papers do not engage meaningfully with children's data privacy and security or notions of children's dataveillance (Clarke, 1988). Indeed, it is questionable whether Internet-connected toys are compliant with the requirements of the Privacy Act 1988, and the Enhancing Online Safety for Children Act 2015.

The Internet of Toys is not merely an expansion of Internet apps and websites for children, or a modest technical extension to existing categories of children's toys. Rather, the move towards the Internet-connected toy cupboard involves a complexity of technological affordances and elements not seen before in the children's consumer market. The growth in this market segment constitutes 'a complex netting of independent but interoperable systems, implemented in a symbiosis with new services and different modes of communication' (Weber, 2009, p. 523). Thus, conventional Internet governance models need to be further developed to accommodate the decentralised architecture involved in the Internet of Toys. Like the Internet itself, IoT architecture (of which the Internet of Toys is a part), continues to evolve from a range of separate contributions. This development:

> involves an increasing number of smart interconnected devices and sensors (e.g. cameras, biometric and medical sensors) that are often non-intrusive, transparent and invisible. Moreover, as the communication among these devices as well as with related services, is expected to happen anytime, anywhere, it is frequently done in a wireless, autonomic and ad-hoc manner. In addition the services become much more fluid, decentralized and complex. Consequently, the security barriers in [the] Internet of Things become much thinner. It also becomes much simpler to collect, store, and search personal information and endanger people's privacy. Finally, concern is rising that control over personal information is increasingly out of the hands of people. Obviously, this goes beyond the risks people are currently used to, leading to new security requirements and to a general requirement /obligation of transparency and accountability when providing IoT services (European Commission, 2013, p. 1–2).

To date, there has been a tendency for developers/designers in the IoT space to concentrate on core technical challenges at the design stage leaving data protection and privacy, as well as information security, for a later development stage. Solutions are often 'added on later once the intended functionality is in place' (European Commission, 2013, p.2). These patch-it-up solutions seem an unsatisfactory way to ensure data minimisation and/or data security and privacy, especially for children. In light of the existing data security failures with Internet-connected toys, more effective regulatory governance, which obliges

designers and developers to incorporate privacy-by-design principles at the foundational stages of product development, would be a preferable solution.

Conclusion and recommendations

Contemporary childhood has undergone a profound transformation, where children's play and social interaction is migrating to the digital realm with a move towards child generated content and child generated communications. Industry has taken note of these changes, creating many opportunities for children to play and socialise online including new, screen-less platforms such as Internet-connected toys. At the same time as industry is creating opportunities for children's cultural participation online, they are, however, gathering vast amounts of user/child information, either surreptitiously or with the consent of parents or children – who recognise that opting out of data collection and sharing practices may diminish the user experience.

There are risks related to the illegitimate access to children's information by hackers and others. Seemingly less worrisome are concerns around corporate surveillance of children's activities and encroachments upon their data privacy and security (or 'children's dataveillance'), along with links to such associated cloud-connected technologies as geo-locational tracking (Wilken & Goggin, 2015). Although commercial data from children's virtual worlds are currently being collected, collated, and passed on to other entities (Grimes, 2008; Steeves & Jones, 2010), the Internet of Toys lifts risk to a whole new level. The range of sophisticated sensors embedded in Internet-connected toys, such as microphones, location and movement detectors, touch sensors and cameras magnify the security and privacy risks of hacking and hijacking. Together with voice recognition software, and a capacity for facial recognition processing, these new dataveillance (Clarke, 1988) capabilities generate huge quantities of information about a child and their social context which can be collected and distributed without the family knowing or understanding the implications. In this sense, therefore, there will be a variety of new (as well as those already known) potential risks as the market develops 'concerning information security and both privacy and data protection, which must be considered' (European Commission, 2103, p.1).

While the emergence of the Internet of Toys raises questions about community expectations regarding the privacy, security, and transparency of Internet-connected toys, research is also needed to investigate the contexts in which such toys are used by children, as well as to analyse the implications for play practices and children's material culture. Obvious also is the need for researchers to assess the emerging risks and benefits associated with children's uses of Internet-connected toys. Such research should include a focus on play within the domestic context of Internet-connected toy use, in order to examine and contextualise the risks, benefits, and changes to children's play culture within the family, and with their friends.

There is currently little reliable information about children's Internet-connected toy use in Australia, even though the subject is gaining media attention (Fritz, 2015; Kobie, 2015; Munro, 2015) and is increasingly the focus of technology bloggers and commentators. Given that consumers are becoming aware of toy data breaches (Office of Children's eSafety Commissioner, 2015a), and policy discussions around the IoT more generally (ACMA, 2015), community concerns can be expected to grow. The Internet of Toys is likely to become an increasingly important topic for public concern and discussion over

the coming years. At the present time it has not been publically addressed in Australian policy circles. Policy development is underway in the broader domain of the IoT in Australia, Europe, and the US, however, and it is to be hoped that further policy statements and initiatives occur. With an increasing demand for specific evidence-based policy, more research is needed in order to inform policy initiatives in the Internet of Toys space.

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